

Pre-Registration Document

Trust-Gated Decision Mechanisms Benchmark

(TGAI-Bench)

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STATUS: Pre-registration complete — no amendments permitted after analysis begins

VERSION: 1.0 (LOCKED)

REPOSITORY:

<https://github.com/IanKubath/trust-gated-decision-benchmark>

LICENSE: MIT

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1 Research Question

1.1 Primary Question

Can a trust-gating function $g(x)$ fitted on one decision domain transfer to structurally different domains with **only unit rescaling** (no functional reshaping)?

1.2 Secondary Question

What functional form (**logistic**, **linear**, **threshold**) best balances performance across heterogeneous domains?

1.3 Rationale

Trust-weighted information combination is well-established in existing literature (power priors, SafeBayes, mixture-of-experts). The novel contribution is the cross-domain empirical validation, not the gating mechanism itself.

2 Hypotheses (Explicitly Falsifiable)

2.1 Main Hypothesis (H_1)

A logistic gating function

$$g(x) = \sigma(a \cdot D(x) - b \cdot S(x) - c)$$

fitted on Domain A will calibrate on Domains B and C within **1.5× calibration error** of domain-specific refit (after only rescaling units).

2.2 Null Hypothesis (H_0)

Domain-specific functional forms outperform transferred forms by **>2.0×**, indicating no transferable shape.

2.3 Competing Prediction

Transfer will succeed for continuous noise domains (Sensor Fusion) but fail for adversarial domains (Fraud), where binary gates are required.

2.4 Failure Condition (Triggers Rejection of H_1)

If transferred calibration error exceeds **2.0×** domain-specific refit on any domain, reject the transfer hypothesis.

3 Experimental Domains

Critical Requirement: At least two domains must use real, independently-collected datasets to avoid “built-to-pass” criticism. Domains B and C satisfy this requirement.

4 Functional Form Specification (Locked Before Fitting)

The gating function $g(x)$ will take one of three forms. The **primary form** is selected below BEFORE any fitting or analysis:

Domain	Data Source	Ground Truth	Corruption Type
A: Sequential Testing	Synthetic (controlled)	(controlled) Coin bias (known)	Prior staleness / drift
B: Sensor Fusion	KITTI Odometry Dataset (REAL)	GPS/IMU trajectory	Gaussian sensor noise
C: Fraud Detection	IEEE-CIS Fraud Dataset (REAL)	Transaction labels	Adversarial framing

4.1 Primary Choice: Logistic Gate

$$g(x) = \frac{1}{1 + e^{-(a \cdot D(x) - b \cdot S(x) - c)}}$$

Parameters: (a, b, c)

Where:

- $D(x)$ = divergence/mismatch statistic
- $S(x)$ = stakes/asymmetry cost
- a, b, c = fitted parameters from Domain A

4.2 Ablation Checks

- **Linear Gate:** $g(x) = \text{clip}(\alpha \cdot D(x) + \beta, 0, 1)$ — Parameters: (α, β)
- **Threshold Gate:** $g(x) = 1$ if $D(x) < \tau$, else 0 — Parameter: (τ)

These ablation forms will **NOT** influence the primary hypothesis test.

5 Success/Failure Criteria (Numeric Thresholds)

Metric	Success Threshold	Failure Threshold
Calibration (ECE)	Error Transferred $\leq 1.5 \times$ domain-specific refit	Transferred $> 2.0 \times$ domain-specific refit
AUC-ROC	Transferred ± 0.05 within of refit	Transferred > 0.10 below refit
Brier Score	Transferred $\leq 1.5 \times$ domain-specific refit	Transferred $> 2.0 \times$ domain-specific refit

Decision Rule: Two out of three metrics must meet success thresholds to claim transfer. If only one metric succeeds: **partial transfer** (document as mixed result). If zero metrics succeed: **complete failure** (reject H_1).

6 Analysis Plan

Step 1: Train on Domain A (Sequential Testing)

- Split data: 70% train, 15% validation, 15% test
- Fit (a, b, c) to minimize calibration error on validation set
- Record optimal parameters
- Store trained gating function

Step 2: Transfer to Domain B (Sensor Fusion)

- **DO NOT** refit parameters
- Only rescale $D(x)$ units to match Domain A range
- Measure calibration error on test set
- Compare against domain-specific refit
- Record ECE, AUC-ROC, Brier Score

Step 3: Transfer to Domain C (Fraud)

- Same procedure as Step 2
- Note: Adversarial domain may fail transfer
- Record ECE, AUC-ROC, Brier Score

Step 4: Ablation Study

- Test Linear and Threshold forms on all three domains
- Determine if functional form matters more than parameters
- Document which form works best per domain

Step 5: Meta-Analysis

- Aggregate results across three domains
- Classify as: Full Transfer / Partial Transfer / Complete Failure
- Write findings for publication

7 Risk Mitigation

8 Expected Outcomes & Interpretation

Note: All three outcomes are **publishable**. Negative results are explicitly welcomed and protected by pre-registration.

Risk	Mitigation Strategy
Dataset unavailable	Backup datasets ready: UCI Credit Card Fraud, NASA Turbopan Degradation
Ground truth mismatch	Pre-process all datasets to ensure binary classification target
Features not comparable	Normalize all $D(x)$ to $[0, 1]$ range using min-max scaling
Overfitting during fit	Use separate validation set for hyperparameter tuning
Reviewer objection to synthetic Domain A	Lead with real datasets; treat Synthetic Domain A as controlled test case

Outcome	Implication
Full Transfer	TGAI is domain-agnostic; validate as general framework. Publish as “Cross-domain validation of trust-gated decision mechanisms.”
Partial Transfer	Works for continuous noise, fails for adversarial domains. Publish as “Domain-specific constraints on trust-gating functions.”
Complete Failure	Binary gates required when signals are unobservable. Publish as “Information deficits prevent continuous trust weighting in deployment.”

Milestone	Target Date
Pre-registration locked	July 29, 2026
Data collection complete	August 5, 2026
Domain A training complete	August 12, 2026
Transfer testing complete	August 20, 2026
Analysis complete	August 27, 2026
Draft methods note	September 10, 2026

9 Timeline

10 Author Commitments

1. **No HARKing:** No changing hypotheses after seeing results. All hypotheses defined in Section 2 are locked before analysis begins.
2. **Transparent Reporting:** Full code and data available on GitHub repository. No selective reporting of successful metrics only.
3. **Negative Results Accepted:** Failure to transfer is a valid contribution. Will report full results regardless of outcome.
4. **Literature Acknowledgement:** Will cite power priors (Ibrahim & Chen, 2000), SafeBayes (Grünwald & van Ommen), and mixture-of-experts (Jacobs et al., 1991) as established frame-

work. This work does NOT claim to invent the gating mechanism, only to empirically test cross-domain transfer.

5. **Data Availability:** Real datasets will be referenced with download links. Synthetic domain code will be reproducible with fixed random seeds.
6. **Version Control:** All changes to this pre-registration must be documented with rationale. Amendments after analysis begins are prohibited.

11 Citation Information

If using this benchmark or citing the pre-registration:

```
@software{kubath2026tgai,
  author = {Kubath, Ian},
  title = {Trust-Gated Decision Mechanisms Benchmark (TGAI-Bench)},
  year = {2026},
  version = {1.0},
  url = {https://github.com/IanKubath/trust-gated-decision-benchmark},
  license = {MIT}
}
```

For the pre-registration document specifically:

```
@misc{kubath2026preregistration,
  author = {Kubath, Ian},
  title = {Pre-Registration: Cross-Domain Transfer Test of Trust-Gated Decision
    Mechanisms},
  year = {2026},
  url = {https://github.com/IanKubath/trust-gated-decision-benchmark/blob/main/PRE_REGISTRATION}
}
```

12 Contact & Version History

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Version History

Version	Date	Changes
1.0	July 29, 2026	Initial pre-registration locked before analysis

NOTE: No amendments permitted after analysis begins without explicit justification and documentation in version history.

END OF PRE-REGISTRATION DOCUMENT